

RICHMOND, VA, USA

WMO: 724010

Lat: 37.512N

Lon: 77.323W

Elev: 164

StdP: 14.61

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13740

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	21.2	-1.1	5.2	23.3	3.5	6.6	26.8	25.3	46.8	23.1	44.2	6.9	0	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	94.9	75.5	92.4	75.0	90.1	74.2	78.4	89.1	77.4	87.6	76.4	85.7	9.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.7	135.3	82.7	74.6	130.4	81.5	73.6	126.0	80.4	42.0	89.4	40.8	87.5	39.8	85.9	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	21.1	18.7	16.7	DB	8.8	98.9	5.8	3.1	4.7	101.1	1.3	102.9	-2.0	104.6	-6.2	106.9
(5)				WB	7.5	80.6	5.4	1.2	3.6	81.5	0.4	82.2	-2.6	82.8	-6.6	83.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	38.5	41.4	48.8	59.1	67.2	75.6	79.3	71.6	60.6	49.9	42.5
		DBStd	16.38	10.68	9.65	9.68	8.56	7.54	5.84	4.57	4.54	6.01	7.93	8.59
		HDD50	1202	384	272	144	17	0	0	0	0	8	106	271
		HDD65	3635	822	661	509	218	68	4	0	14	184	456	699
		CDD50	4654	28	32	106	290	534	767	909	862	648	336	104
		CDD65	1609	0	0	6	41	137	321	444	397	212	47	3
		CDH74	14926	1	6	74	450	1226	3054	4481	3669	1583	359	18
		CDH80	5984	0	0	10	127	399	1264	2039	1538	522	84	1
(14)	Wind	WSAvg	7.4	8.1	8.3	8.9	8.9	7.7	7.1	6.6	6.1	6.6	6.7	7.1
(15)	Precipitation	PrecAvg	44.0	3.2	2.9	3.8	3.0	3.9	4.1	4.8	4.6	4.0	3.4	3.3
(16)		PrecMax	63.7	8.0	6.0	8.7	8.3	10.4	13.3	13.9	16.3	16.6	9.4	9.6
(17)		PrecMin	28.3	0.6	0.5	0.2	0.6	0.9	0.4	0.5	0.6	0.1	0.0	0.2
(18)		PrecStd	8.5	1.5	1.3	1.9	1.5	2.0	2.6	2.6	2.8	3.2	2.2	2.0
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	71.4	73.8	81.3	88.3	91.4	97.1	99.1	96.8	93.0	87.2	76.6	73.0
(20)		0.4% MCWB	61.1	61.4	62.3	67.3	72.2	75.1	76.6	76.5	73.5	70.8	64.4	62.1
(21)		2% DB	65.7	67.8	75.3	83.9	87.7	93.1	95.1	93.8	89.3	81.9	72.3	67.0
(22)		2% MCWB	59.2	56.9	59.8	64.8	71.3	74.7	76.3	75.4	72.7	68.9	62.0	61.1
(23)		5% DB	61.1	62.6	70.4	79.5	84.7	90.2	92.6	90.9	85.9	77.9	68.6	62.3
(24)		5% MCWB	55.0	53.6	57.1	63.1	69.9	73.6	75.7	74.9	71.3	67.0	60.2	56.7
(25)		10% DB	55.7	57.8	65.5	74.8	81.2	87.2	90.0	88.1	82.6	74.1	64.7	57.4
(26)		10% MCWB	49.3	49.6	54.9	61.4	68.0	72.6	75.3	73.9	70.2	65.2	57.8	51.6
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	64.4	64.2	65.6	71.2	75.6	78.1	80.4	80.1	77.4	75.2	68.5	65.9
(28)		0.4% MCDB	67.9	71.2	75.6	80.4	85.9	89.7	92.0	90.7	85.1	79.8	72.3	69.5
(29)		2% WB	60.4	59.9	62.8	68.3	73.4	76.7	78.9	78.3	75.9	72.2	65.2	61.9
(30)		2% MCDB	64.9	65.0	71.5	78.7	83.7	87.7	90.0	88.5	83.4	78.1	69.8	66.2
(31)		5% WB	56.0	55.4	60.1	65.9	71.7	75.7	77.8	77.1	74.4	69.5	62.1	57.7
(32)		5% MCDB	59.8	60.7	67.7	75.4	81.4	86.3	88.5	86.6	81.1	76.1	66.9	61.1
(33)		10% WB	50.2	50.5	56.7	63.7	70.0	74.6	76.6	76.0	72.9	66.7	59.0	52.7
(34)		10% MCDB	54.5	56.9	63.8	72.2	78.5	84.2	86.4	84.2	79.0	72.7	63.7	56.2
(35)	Mean Daily Temperature Range	MDBR	17.3	19.3	20.7	22.1	20.1	19.1	18.6	18.3	18.3	19.9	19.8	17.5
(36)		5% DB MCDBR	23.9	26.2	27.1	27.4	23.4	22.2	21.8	21.6	22.0	22.9	23.9	22.7
(37)		5% WB MCWBR	18.3	18.0	15.1	12.5	10.5	8.0	7.0	7.0	8.5	11.7	14.8	17.2
(38)		5% WB MCDBR	22.0	23.6	23.8	23.3	20.3	19.6	19.2	18.0	16.8	18.9	19.0	20.2
(39)		5% WB MCWBR	18.9	19.0	16.1	12.8	9.7	8.1	7.3	7.3	8.0	11.1	14.4	17.4
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.314	0.346	0.390	0.427	0.486	0.505	0.506	0.407	0.353	0.320	0.304
(41)		taud	2.532	2.484	2.432	2.325	2.246	2.127	2.095	2.076	2.339	2.490	2.537	2.556
(42)		Ebn at Noon	283	292	291	283	272	255	249	246	267	274	273	274
(43)		Edh at Noon	24	29	34	40	44	49	50	50	36	29	24	23
(44)	All-Sky Solar Radiation	RadAvg	733	974	1290	1625	1846	2005	1911	1726	1406	1101	827	646
(45)		RadStd	61	108	105	126	170	138	96	83	145	131	66	56

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+290	+193
(47)	Regional (72 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+261	+164

Nomenclature: See separate page